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Quantum Computing

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1

Complex linear algebra

1.1 Complex numbers

Complex number

A **Complex number** z is defined as:

$$z = a + ib$$

where $b \in \mathbb{R}$.

From a complex number we can define:

- Coniugate: $\overline{z} = \overline{a + ib} = a - ib$
- Modulus: $|z| = \sqrt{a^2 + b^2}$

Given a number z and its modulus $|z|$, we can find an angle $\theta \in [-\pi, \pi]$ such that:

$$\cos\theta = \frac{a}{\sqrt{a^2 + b^2}} = \frac{a}{|z|}$$
$$\sin\theta = \frac{b}{\sqrt{a^2 + b^2}} = \frac{b}{|z|}$$

So, we can write:

$$z = |z|(\cos\theta + i\sin\theta)$$

We also define the exponential of an imaginary number:

$$e^{i\theta} = \cos\theta + i\sin\theta \implies z = |z|e^{i\theta}$$

1.2 Vectors and Matrices

A vector n is list of complex numbers, for which we can define a norm:

$$||v|| = \sqrt{|v_1|^2 + \cdots + |v_n|^2} = \sqrt{\sum_{i=1}^{n} |v_i|^2}$$

Example:
Given the vector:

$$v = [4, i, 2 + 3i]$$

Its norm is:

$$||v|| = \sqrt{16 + 1 + 13} = \sqrt{30}$$

A matrix is a table of complex number, for which the multiplication is not commutative. The multiplication between a matrix A with size $m \times n$ and a matrix B with size $n \times q$ has size $m \times q$. The result matrix C has in every cell:

$$C_{i,j} = \sum_{k=0}^{n} A_{i,k} \cdot B_{k,j}$$

Example:
Given the matrices:

$$A = \begin{bmatrix} 3 & i \\ 1+i & 0 \\ -4 & 2+3i \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix}$$

The multiplication $A \cdot B$ is:

$$C = \begin{bmatrix} 9 & 3 \\ 3+3i & 1+i \\ -12 & -4 \end{bmatrix}$$

1.2.1 Eigenvectors and Eigenvalues

A vector v is an **eigenvector** with **eigenvalue** λ for a matrix A if:

$$A \cdot v = \lambda \cdot v$$

We can also write:

$$A \cdot v - \lambda \cdot v = 0 \iff (A - \lambda I) \cdot v = 0$$

This equation has non-zero solutions only if:

$$\det(A - \lambda I) = 0$$

In general v and λ could be complex, but quantum mechanics uses matrices that always have eigenvalues $\lambda \in \mathbb{R}$.

2

Introduction

2.1 Probabilistic algorithms

In normal programming, an algorithms produce a fixed output for a given input. With probabilistic algorithms there is a finite set of possible outputs $\{o_1, \ldots, o_N\}$ for a given input. Each outputs o_i has a probability of p_i to be produced, where:

$$\sum_{k=1}^{N} p_k = 1$$

Classical algorithms can have probabilistic outputs based on pseudo-random number generator (PRNG), that produce a sequance of numbers that appear random but after many elements the sequence repeats. In quantum algorithms there is no PRNG, but real randomicity.

2.2 Qubit

Qubit

A **qubit** is the quantum equivalent of a bit, where the basis states are:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The true state of a qubit ψ is a **superposition** defined as:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{C}$ are called **probability amplitudes** and $|\alpha|^2 + |\beta|^2 = 1$.

The notation $|\psi\rangle$ is called "ket".

Example:

Given the vector:

$$v = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

The corresponding qubit is:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} \left(|0\rangle + |1\rangle \right)$$

The true state of a qubit cannot be observed, but only measured. Measuring a qubit makes it collapse into $|0\rangle$ or $ket1$ with the probabilities $|\alpha|^2$ and $|\beta|^2$. We write this property as:

$$\alpha\,|0\rangle + \beta\,|1\rangle \begin{cases} |0\rangle & @\ |\alpha|^2 \\ |1\rangle & @\ |\beta|^2 \end{cases}$$

where @ is called "at".
For any complex number z such that $|z| = 1$, we have that $|z\alpha| = |\alpha|$, so when measuring we have that:

$$z\alpha\,|0\rangle + z\beta\,|1\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

because the probability is based on the amplitudes squared.

2.2.1 Bloch sphere

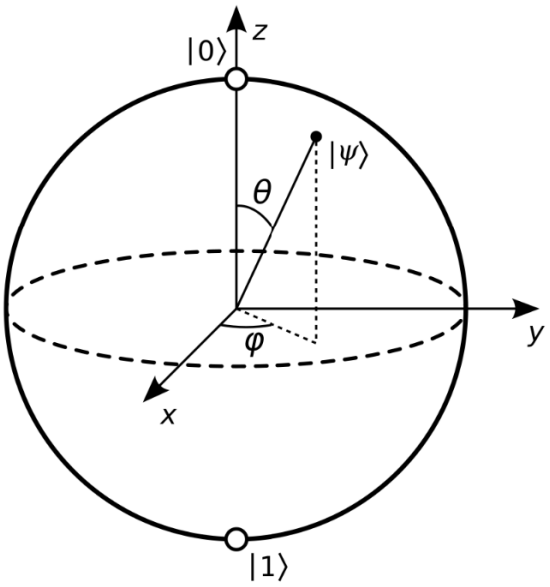
When can picture a qubit using the **Bloch sphere**. Considering a qubit:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

Rewriting it using polar coordinates:

$$|\psi\rangle = e^{i\gamma}\left(\cos\frac{\theta}{2}\,|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}\,|1\rangle\right)$$

Since $|e^{i\gamma}| = 1$ we can remove it, representing the qubit based on two angles $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi)$.



There is no Bloch sphere equivalent for systems composed by two or more qubits.

2.3 Qubit operations

Qubits transformations are linear, so they can be represented as a matrix.

Pauli matrices

The **Pauli matrices** are 2×2 matrices that represent operations:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

These are called the X, Y and Z gate respectively.

The result of applying this gates on the two basis states $|0\rangle$ and $|1\rangle$ are:

- X gate (also called NOT):

$$\sigma_x |0\rangle = |1\rangle \qquad \sigma_x |1\rangle = |0\rangle$$

- Y gate:

$$\sigma_y |0\rangle = i |1\rangle \qquad \sigma_y |1\rangle = -i |0\rangle$$

- Z gate:

$$\sigma_z |0\rangle = |0\rangle \qquad \sigma_z |1\rangle = - |1\rangle$$

Another important transformation is the **Hadamard gate**, represented by the following matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

This gate can map a classical state into a superposition:

$$H |0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

If we measure it we will have a true random bits generator:

$$\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \begin{cases} |0\rangle & @ \ \frac{1}{2} \\ |1\rangle & @ \ \frac{1}{2} \end{cases}$$

We define the result of applying the Hadamard gate on the two classical state with special names:

$$H |0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+\rangle \qquad H |1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |-\rangle$$

To be an allowed quantum transformation a matrix must preserve the norm of the input vector, so it must be **unitary**. Since $|\alpha|^2 + |\beta|^2 = 1$, the norm of every vector v must be $||v|| = 1$ for any qubit state. An unitary matrix preserve the probabilities and is reversible, so from the output we can recover the input.

2.4 Multiple qubits

In classical computers we can describe n bits using an vector of elements and all the possible combination are obtained by performing the cartesian product:

$$\mathcal{B} = \{0, 1\} \implies \mathcal{B} \times \mathcal{B} = \{00, 01, 10, 11\}$$

The lenght of the vector grows linear in respect to n and each element is independent from the others.
To represent all the state of two qubits we need to use the **tensor product**. Given two qubits, the tensor product of their state vectors:

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix}\alpha_0\\ \alpha_1\end{pmatrix} \otimes \begin{pmatrix}\beta_0\\ \beta_1\end{pmatrix} = \begin{pmatrix}\alpha_0\beta_0\\ \alpha_0\beta_1\\ \alpha_1\beta_0\\ \alpha_1\beta_1\end{pmatrix} = \begin{pmatrix}\alpha_0 |\phi\rangle\\ \alpha_1 |\phi\rangle\end{pmatrix} = |\psi\phi\rangle$$

This product can also be written by using the **diagonal (or computational) basis vectors**, that are vectors with all 0 and a 1 in a single position i, called e_i.

$$\alpha_0\beta_0 \underbrace{\begin{pmatrix}1\\0\\0\\0\end{pmatrix}}_{e_0} + \alpha_0\beta_1 \underbrace{\begin{pmatrix}0\\1\\0\\0\end{pmatrix}}_{e_1} + \alpha_1\beta_0 \underbrace{\begin{pmatrix}0\\0\\1\\0\end{pmatrix}}_{e_2} + \alpha_1\beta_1 \underbrace{\begin{pmatrix}0\\0\\0\\1\end{pmatrix}}_{e_3} = \alpha_0\beta_0 |00\rangle + \alpha_0\beta_1 |01\rangle + \alpha_1\beta_0 |10\rangle + \alpha_1\beta_1 |11\rangle$$

We can see that each vector e_i is equal to the vector state of $|i\rangle$ in which we consider the binary representation of i. So, the previous result can also be written as a sum:

$$\sum_{i,j\in\{0,1\}} \alpha_i\beta_j |ij\rangle$$

Example:
Given two qubits:

$$|\psi\rangle = |\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Their tensor product is:

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{pmatrix} \otimes \begin{pmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{pmatrix} = \begin{pmatrix}\frac{1}{2}\\ \frac{1}{2}\\ \frac{1}{2}\\ \frac{1}{2}\end{pmatrix} = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

2.4.1 Entanglement

In the example above the two qubits can be separated and decomposed into two valid qubit states.
Since any qubit written:

$$\alpha_0 |00\rangle + \alpha_1 |01\rangle + \alpha_2 |10\rangle + \alpha_3 |11\rangle \quad \text{with } \sum_i |\alpha_i|^2 = 2$$

is a valid quantum state for a 2-qubit system we can have a valid qubit such as:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

It's not possible to separate this state in two different quantum states for which their tensor product is $|\psi\rangle$. This would require:

$$|a\rangle \otimes |b\rangle = \alpha_0\beta_0 |00\rangle + \alpha_0 v\beta_1 |01\rangle + \alpha_1\beta_0 |10\rangle + \alpha_1\beta_1 |11\rangle = \begin{cases} \alpha_0\beta_0 = \alpha_1\beta_1 = 0 \\ \alpha_0\beta_1 = \alpha_1\beta_0 = \frac{1}{\sqrt{2}} \end{cases}$$

The product $\alpha_0\beta_0 = 0$ implies that at least one of the two is 0, but this means also that at least one of $\alpha_0\beta_1$ or $\alpha_1\beta_0$ is 0.
This state is **entangled**, it means that it doesn't describe two qubit states but the entire 2-qubit system as a whole. This entangled states are called the **EPR** states and when measuring them we see that:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) \begin{cases} |01\rangle & @ \frac{1}{2} \\ |10\rangle & @ \frac{1}{2} \end{cases}$$

This pairs also have a strange behaviour when physically separated. Since we cannot measure $|00\rangle$ or $|11\rangle$, measuring one of the two qubit is enough to know the entire system, also when the two qubits are kilometers apart. In fact is proved that if we take the two qubits apart and measure one of the two, the other will also collapse its state.

2.4.2 Mutiple qubits gates

The basic gate for two qubits its the CNOT gate, where the first bit is the **control** and the second one the **target**. The target bit is flipped when the control is 1, so the truth table is:

a	b	CNOT(a,b)
0	0	0
0	1	1
1	0	1
1	1	0

Logically we have that:

$$\text{CNOT}(a, b) = a \oplus b$$

Since we want the computation to be invertible we need the value of a to recover the value of b, so the quantum CNOT behaves as follows:

$$\text{CNOT}(|a\rangle \otimes |b\rangle) = |a\rangle \otimes |a \oplus b\rangle = |a, a \oplus b\rangle$$

The matrix that represent this gate is:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

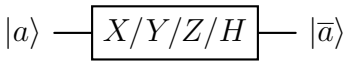
2.5 Quantum circuit

In quantum computing we program using gates and circuits. Each circuit U must be unitary, that means:

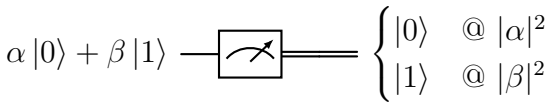
$$UU^\dagger = U^\dagger U = I$$

Where $U^\dagger$ is the **adjoint** of U, given by $U^\dagger_{i,j} = \overline{U_{j,i}}$.
The notation used for the gates seen before is:

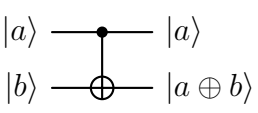
- X, Y, Z and H gates: single bit with the correspondig letter

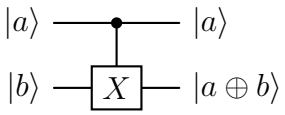


- Measurment gate: the double wire after means that is passing classical bits, not quantun ones



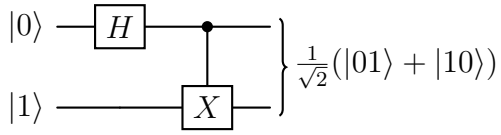
- CNOT gate: the wire with the dot is the control and the one with the gato or xor is the target





2.5.1 Creating EPR pairs

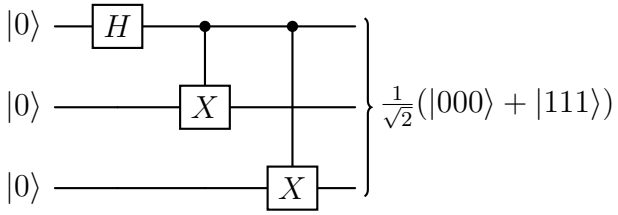
There are different circuits capable of creating EPR pairs, one of the common is composed by H and CNOT gates.



We can evaluate the circuit by sequentially crossing the gates from left to right:

$$|0\rangle \otimes |1\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

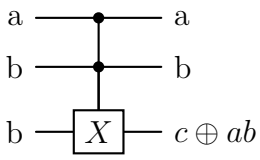
Another gate that creates EPR pairs, but with 3 qubits is the GHZ (Greenberger, Horne, Zeilinger) gate:



Tecnically all quantum circuits starts with all $|0\rangle$ as inputs, but we can simply add a NOT gate to change it to $|1\rangle$.

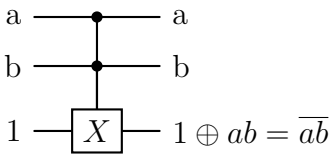
2.5.2 Reversible classical computing

Using a specific quantum gate, called the **Toffoli** gate (also called CCNOT), we make classical computing reversible.

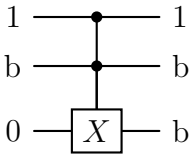


With this gate we can create the NAND and FANOUT gate, that means that any boolean circuit can be implemented only with Toffoli gate, making it reversible.

NAND gate



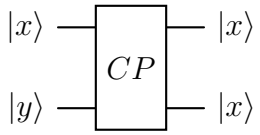
FANOUT gate



2.6 Cloning and Teleportation

2.6.1 No cloning theorem

A copy operation takes a source variable to be copied and a destination variable where the copy is applied. As a circuit we could write:



We need a variable $|y\rangle$ that is a "white paper" where the content of $|x\rangle$ can be copied, but this cannot exists.

No cloning theorem

There is no operation CP, such that:

$$\exists\, |y\rangle\, |\forall\, |x\rangle \;\; \mathrm{CP}(|x\rangle \otimes |y\rangle) = |x\rangle \otimes |x\rangle$$

Proof by contradiction:

For the proof the $|\rangle$ notation will be avoided.

CP must work on superposed states, so:

$$\exists y | \forall x, a \;\; \mathrm{CP}(\underbrace{(x+a) \otimes y}_{(x \otimes y)+(a \otimes y)}) = (x+a) \otimes (x+a)$$

CP must be linear:

$$\exists y | \forall x, a \ \mathrm{CP}(x \otimes y) + \mathrm{CP}(a \otimes y) = (x + a) \otimes (x + a)$$

Since CP copies its input:

$$\forall x, a \ (x \otimes x) + (a \otimes a) = (x + a) \otimes (x + a) = (x \otimes x) + (x \otimes a) + (a \otimes x) + (a \otimes a)$$

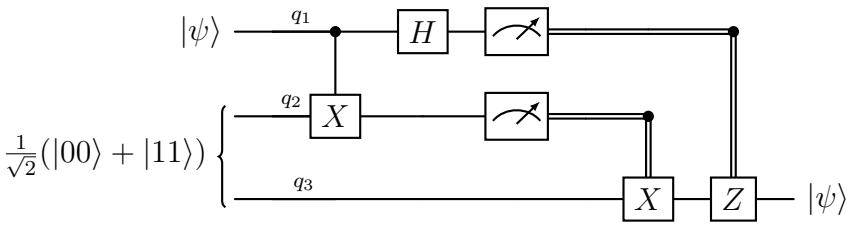
But we can prove with a simple example like $x = |0\rangle$ and $a = |1\rangle$ that this equation is not true.

$$\left.\begin{array}{l} x = |0\rangle \\ a = |1\rangle \end{array}\right\} \implies (x \otimes x) + (a \otimes a) = |00\rangle + |11\rangle \neq (x \otimes x) + (x \otimes a) + (a \otimes x) + (a \otimes a) = |00\rangle + |01\rangle + |10\rangle + |11\rangle$$

This means that we cannot clone perfectly, but **imperfect cloning** with high probability is possible.

2.6.2 Quantum teleportation

Even if we cannot perfectly clone a qubit, we can teleport it perfectly fine.
Suppose Alice has a qubit $|\psi\rangle$ and she wants to send it to Bob, this can be achieved by using 2 qubits and 2 classical bits. The Quantum teleportation circuit is the following:



We can give a pseudocode for the algorithm:

Algorithm: Quantum Teleportation

```
def Teleport(q1, q2, q3):
    q1, q2, q3=|ψ⟩ 1/√2(|00⟩ + |11⟩)
    q1, q2=CNOT(q1, q2)
    q1=H(q1)
    b1, b2=Measure(q1, q2)
    b2, q3=CNOT(b2, q3)
    b1, q3=CZ(b1, q3)
    return q3
```

We can prove that the algorithm works by executing the operations done by the code step by step, starting from a random qubit ψ.

$$|\psi\rangle \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (\alpha |0\rangle + \beta |1\rangle)\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}} [\alpha |0\rangle (|00\rangle + |11\rangle) + \beta |1\rangle (|00\rangle + |11\rangle)]$$

Applying the CNOT on q_1 and q_2 we obtain:

$$\frac{1}{\sqrt{2}} [\alpha |0\rangle (|00\rangle + |11\rangle) + \beta |1\rangle (|10\rangle + |01\rangle)]$$

Next, we apply the Hadamard gate on q_1:

$$\frac{1}{\sqrt{2}}\left[\alpha\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)(|00\rangle+|11\rangle)+\beta\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)(|10\rangle+|01\rangle)\right]=$$
$$\frac{1}{2}\left[\alpha(|000\rangle+|011\rangle+|100\rangle+|111\rangle)+\beta(|010\rangle+|001\rangle-|110\rangle-|101\rangle)\right]=$$
$$\frac{1}{2}\left[|00\rangle\left(\alpha|0\rangle+\beta|1\rangle\right)+|01\rangle\left(\alpha|1\rangle+\beta|0\rangle\right)+|10\rangle\left(\alpha|0\rangle-\beta|1\rangle\right)+|11\rangle\left(\alpha|1\rangle-\beta|0\rangle\right)\right]$$

Now we have four possible outcomes based on the measurment done by Alice. Given the measured bits Bob need to apply different operation on its remaining qubit, to obtain the initial qubit $|\psi\rangle\ (\alpha|0\rangle+\beta|1\rangle)$.

Alice bits	Bob qubit	Operations
00	$\|\phi\rangle=\alpha\|0\rangle+\beta\|1\rangle$	$I\|\phi\rangle=\|\psi\rangle$
01	$\|\phi\rangle=\alpha\|1\rangle+\beta\|0\rangle$	$X\|\phi\rangle=\|\psi\rangle$
10	$\|\phi\rangle=\alpha\|0\rangle-\beta\|1\rangle$	$Z\|\phi\rangle=\|\psi\rangle$
11	$\|\phi\rangle=\alpha\|1\rangle-\beta\|0\rangle$	$XZ\|\phi\rangle=\|\psi\rangle$

Since Alice has measured her qubit, the qubit collapsed and she lost it forever, so now only Bob has that qubit.

3

Quantum parallelism

Given any boolean function $f : \mathcal{B}^n \to \mathcal{B}$, we can show that the function:

$$U_f = (x, y) \in (\mathcal{B}^n, \mathcal{B}) \to (x, y \oplus f(x))$$

Is reversible and unitary, so its a valid quantum gate.

Example:

For a 1-bit function we have the truth table:

		$y \oplus f(x)$		
x	y	x	$f(x) = 0$	$f(x) = 1$
0	0	0	0	1
0	1	0	1	0
1	0	1	0	1
1	1	1	1	0

So, we have that:

$$\begin{cases} U_f(x, 0) = (x, f(x)) \\ U_f(x, 1) = (x, \neg f(x)) \end{cases}$$

Since the function U_f can be implemented unitarly, we can apply it to qubits, obtaining:

$$\begin{cases} U_f \left| x \otimes 0 \right\rangle = \left| x \otimes f(x) \right\rangle \\ U_f \left| x \otimes 1 \right\rangle = \left| x \otimes \neg f(x) \right\rangle \end{cases}$$

Example:

Given $\left| x \right\rangle, \left| y \right\rangle$ as:

$$\left| x \right\rangle = \frac{1}{\sqrt{2}}(\left| 0 \right\rangle + \left| 1 \right\rangle) \qquad \left| y \right\rangle = \left| 0 \right\rangle$$

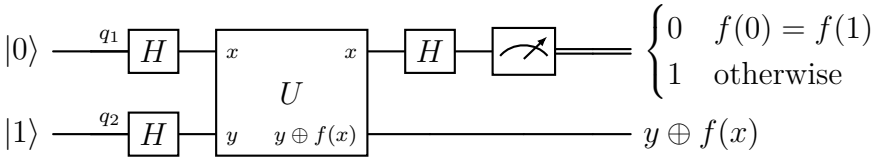
We can apply U_f to obtain:

$$U_f \left(\frac{1}{\sqrt{2}}(\left| 0 \right\rangle + \left| 1 \right\rangle) \otimes \left| 0 \right\rangle \right) = \frac{1}{\sqrt{2}}(U_f \left| 00 \right\rangle + U_f \left| 10 \right\rangle) = \frac{1}{\sqrt{2}}(\left| 0 \otimes f(0) \right\rangle + \left| 1 \right\rangle \otimes f(1))$$

So $f(0)$ and $f(1)$ are computed in **parallel**.

3.1 Deutsch algorithm

Given a bolean function $f : \mathcal{B} \to \mathcal{B}$ we want to compute the value of $f(0) \oplus f(1)$, to know if $f(0)$ is equal to $f(1)$. In classical computing we need 2 evaluation of the function f, but in quantum we can use the **Deutsch algorithm** that needs only one evaluation.



The code for the algorithm is the following:

Algorithm: Deutsch algorithm

```
def Deutsch(f, q1, q2):
    q1, q2=|01⟩
    q1, q2=H⊕H(q1, q2)
    q1, q2=Uf(q1, q2)
    q1=H(q1)
    b=Measure(q1)
    return b
```

Before proving the algorithm we need to know how U_f works on a random state a.

Application of U_f

Given a boolean function f, for any basis state $a \in \mathcal{B}^n$, we have that:

$$U_f\left(|a\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right) = (-1)^{f(a)}|a\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = (-1)^{f(a)}|a-\rangle$$

Proof:

$$U_f\left(|a\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right) = \frac{1}{\sqrt{2}}U_f(|a0\rangle - |a1\rangle) = \frac{1}{\sqrt{2}}(|af(a)\rangle - |a\neg f(a)\rangle)$$

We need to divide the result based on the value of $f(a)$:

$$\frac{1}{\sqrt{2}}(|af(a)\rangle - |a\neg f(a)\rangle) = \begin{cases} \frac{1}{\sqrt{2}}(|a0\rangle - |a1\rangle) & f(a) = 0 \\ \frac{1}{\sqrt{2}}(|a1\rangle - |a0\rangle) & f(a) = 1 \end{cases} = \frac{1}{\sqrt{2}}(-1)^{f(a)}|a\rangle\,(|0\rangle - |1\rangle) = (-1)^{f(a)}|a-\rangle$$

Now, from the starting state $|01\rangle$ we can execute the operations of the algorithm in order.

$$|01\rangle \xrightarrow{H\otimes H(q_1,q_2)} |+-\rangle = \frac{1}{\sqrt{2}}|0-\rangle + \frac{1}{\sqrt{2}}|1-\rangle \xrightarrow{U_f(q_1,q_2)} \frac{1}{\sqrt{2}}\left((-1)^{f(0)}|0-\rangle + (-1)^{f(1)}|1-\rangle\right)$$

Based on the value of $f(0)$ and $f(1)$ we can separate the result:

$$\begin{cases} \frac{1}{\sqrt{2}}(|0-\rangle + |1-\rangle) & f(0)=0, f(1)=0 \\ \frac{1}{\sqrt{2}}(-|0-\rangle - |1-\rangle) & f(0)=1, f(1)=1 \\ \frac{1}{\sqrt{2}}(|0-\rangle - |1-\rangle) & f(0)=0, f(1)=1 \\ \frac{1}{\sqrt{2}}(-|0-\rangle + |1-\rangle) & f(0)=1, f(1)=0 \end{cases} \implies \begin{cases} \pm\frac{1}{\sqrt{2}}(|0-\rangle + |1-\rangle) & f(0)=f(1) \\ \pm\frac{1}{\sqrt{2}}(|0-\rangle - |1-\rangle) & f(0)\neq f(1) \end{cases}$$

$$\implies \begin{cases} \pm\frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle) + \frac{1}{\sqrt{2}}(|10\rangle - |11\rangle)\right) = \pm|+-\rangle \\ \pm\frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle) + \frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)\right) = \pm|--\rangle \end{cases}$$

Now we apply the Hadamard gate only on the first qubit:

$$\begin{cases} \pm\frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle) + \frac{1}{\sqrt{2}}(|10\rangle - |11\rangle)\right) = \pm|+-\rangle \\ \pm\frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle) + \frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)\right) = \pm|--\rangle \end{cases} \implies \begin{cases} |0-\rangle & f(0)=f(1) \\ |1-\rangle & f(0)\neq f(1) \end{cases}$$

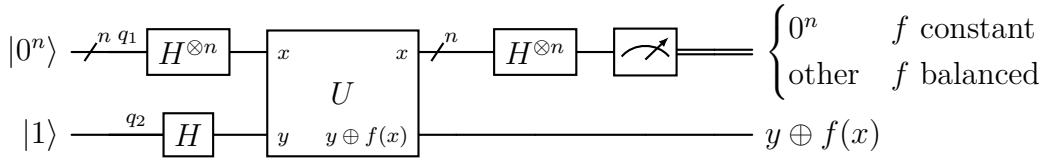
Measuring the first qubit, we will get 0 if $f(0)=f(1)$ and 1 if $f(0)\neq f(1)$, knowing also the result of $f(0)\oplus f(1)$.

3.2 Deutsch-Jozsa algorithm

Any boolean function $f:\mathcal{B}^n\to\mathcal{B}$ can be:

- Constant:
$$\exists b\in\mathcal{B}|\forall x\ f(x)=b$$
- Balanced:
$$\sum_{x\in\mathcal{B}^n} f(x) = \frac{2^n}{2} = 2^{n-1}$$

Given a function f that its either constant or balanced, we want to know wheter f is constant or balanced. The problem in classical computing needs at most $2^{n-1}+1$ evaluation of the function f, but in quantum the computation can be done with only one evaluation, having an exponential speedup.



The code for the algorithm is the following:

Algorithm: Deutsch-Jozsa algorithm

```
def Deutsch_Jozsa(f, q1, q2):
    q1, q2=|0^n 1⟩
    q1, q2=H^⊗n ⊕ H(q1, q2)
    q1, q2=U_f(q1, q2)
    q1=H^⊗n(q1)
    b=Measure(q1)
    return b
```

Also in this case before proving the algorithm we need to know how a gate works, specifically how $H^{\otimes n}$ works.

Application of $H^{\otimes n}$

For any $x \in \mathcal{B}^n$, we have that:

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a} |a\rangle$$

Where $x \cdot a$ is defined as:

$$x \cdot a = \left(\sum_{i=1}^{n} x_i a_i \right) \mod 2$$

Proof:

The result of a single Hadamard gate can be written as:

$$\left.\begin{aligned} H|0\rangle = |+\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \tfrac{1}{\sqrt{2}} \sum_{b \in \mathcal{B}} \overbrace{(-1)^{0 \cdot b}}^{1} |b\rangle \\ H|1\rangle = |-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \tfrac{1}{\sqrt{2}} \sum_{b \in \mathcal{B}} (-1)^{1 \cdot b} |b\rangle \end{aligned}\right\} \implies \forall a \in \mathcal{B}\ H|a\rangle = \frac{1}{\sqrt{2}} \sum_{b \in \mathcal{B}} (-1)^{a \cdot b} |b\rangle$$

So, the tensor product of n Hadamard gate for $x \in \mathcal{B}^n$ will be:

$$\begin{aligned} H^{\otimes n} |x\rangle = \bigotimes_{i=1}^{n} H |x_i\rangle = \bigotimes_{i=1}^{n} \left(\frac{1}{\sqrt{2}} \sum_{b \in \mathcal{B}} (-1)^{x_i \cdot b} |b\rangle \right) = \frac{1}{\sqrt{2^n}} \bigotimes_{i=1}^{n} (|0\rangle + (-1)^{x_i} |1\rangle) \\ = \frac{1}{\sqrt{2^n}} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a} |a\rangle \end{aligned}$$

Now we can follow the instruction of the algorithm starting with the state $|0^n 1\rangle$.

$$|0^n 1\rangle \xrightarrow{H^{\otimes n} H(q_1, q_2)} \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \mathcal{B}^n} \overbrace{(-1)^{0^n \cdot x}}^{1} |x\rangle \right) \otimes |-\rangle \xrightarrow{U_f(q_1, q_2)} \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)} |x\rangle \right) \otimes |-\rangle$$

After applying U_f the second qubit is never used anymore, so we can erase it and continue the computation with only q_1.

$$\begin{aligned} \frac{1}{\sqrt{2^n}} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)} |x\rangle \xrightarrow{H^{\otimes n}(q_1)} \frac{1}{\sqrt{2^n}} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)} \left(\frac{1}{\sqrt{2^n}} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a} |a\rangle \right) \\ = \frac{1}{2^n} \sum_{x \in \mathcal{B}^n} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a + f(x)} |a\rangle \end{aligned}$$

The two sum are not dependant one to the other so we can switch the order:

$$\frac{1}{2^n} \sum_{x \in \mathcal{B}^n} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a + f(x)} |a\rangle = \frac{1}{2^n} \sum_{a \in \mathcal{B}^n} |a\rangle \sum_{x \in \mathcal{B}^n} (-1)^{x \cdot a + f(x)}$$

When measuring q_1 the probability of getting 0^n is computed by setting $a = 0^n$ in the formula above.

$$\frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (-1)^{\overbrace{x \cdot a}^{0} + f(x)} = \frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)}$$

This means that wheter f is constant or balanced we will obtain a precise result:

- Constant:
$$\frac{1}{2^n}\sum_{x\in\mathcal{B}^n}(-1)^{f(x)} = \pm 1$$
- Balanced:
$$\frac{1}{2^n}\sum_{x\in\mathcal{B}^n}(-1)^{f(x)} = 0$$

Finally, with only one evaluation of f we know if it's constant or balanced.

4

Mathematical foundations of quantum

4.1 Vector spaces and Linear maps

Groups

A group is a set G with a multiplication operation that satisfies:

1. Associativity: $a(bc) = (ab)c$
2. Unit element: $\exists 1 \in G | \forall a \in G \; a1 = 1a = a$
3. Inverse: $\forall a \in G \; \exists a^{-1} | aa^{-1} = a^{-1}a = 1$

If the multiplication is commutative the group is called **abelian** group.

Vector spaces

A vector space is a set V with a vector sum and a scalar multiplication that is commutative, making V an abelian group.
The vector sum satisfies:

1. Associativity: $u + (v + w) = (u + w) + v$
2. Null vector: $\exists \overrightarrow{0} | \forall v \in V \; \overrightarrow{0} + v = v + \overrightarrow{0}$
3. Inverse: $\forall v \in V \; \exists -v | v + (-v) = \overrightarrow{0}$
4. Commutative

The scalar multiplication satisfies:

1. $\alpha(u + v) = \alpha u + \alpha v$
2. $(\alpha + \beta)v = \alpha v + \beta v$
3. $\alpha(\beta v) = (\alpha\beta)v$
4. $1v = v$
5. $0v = \overrightarrow{0}$

Linear map

A linear map between vector spaces V and W is a function $L : V \to W$ that satisfies:

$$L(\alpha u + \beta v) = \alpha L(u) + \beta L(v)$$

Given a vector space V, its **dual** V^* is the set of all linear maps $L : V \to \mathbb{C}$.

Inside a vector space we can define a **linear subspace** $W \subset V$ if vector sum and scalar multiplication are closed in W.

Dimension of a vector space

A set of vectors $v_1, \ldots, v_n \in V$ is linearly dependent if:

$$\exists(\alpha_1, \ldots, \alpha_n) \neq (0, \ldots, 0) | \sum_{i=1}^{n} \alpha_i v_i = 0$$

Otherwise the vectors are **linearly independent**.
A vector space is n-dimensional if it has a set of maximum n linearly independent vectors. This set of vectors is called a **basis set** for V.

Given a subspace $S \subset V$, the linear span of S, called $[S]$ is the set of all linear combinations of the vectors in S.

$$[S] = \left\{ \sum_{i=1}^{n} \alpha_i s_i \middle| s_i \in S, \alpha_i \in \mathbb{C}, n \in \mathbb{N} \right\}$$

Given the basis set $S = \{e_1, \ldots, e_n\}$ of a vector space S we have that $[S] = V$, so:

$$\forall v \in V \ \exists \alpha_1, \ldots, \alpha_n | v = \sum_{i=1}^{n} \alpha_i e_i$$

The coefficients α_i are unique for a basis set.

4.2 Scalar inner product

Scalar product

A scalar product over a vector space V is a function $\langle v, w\rangle : V \times V \to \mathbb{C}$ that maps $v, w \in V$ to a complex $\langle v, w\rangle$. The product satisfies:

1. $\langle u, \alpha v + \beta w\rangle = \alpha\langle u, v\rangle + \beta\langle u, w\rangle$
2. $\langle u, v\rangle = \overline{\langle v, u\rangle}$
3. $\langle u, v\rangle \geq 0$ and $\langle u, u\rangle = 0 \iff u = 0$

Given two vectors $u = (u_1, \ldots, u_n), v = (v_1, \ldots, v_n) \in \mathbb{C}^n$ the scalar product is defined as:

$$\langle u, v\rangle = \sum_{i=1}^{n} \overline{u_i} v_i$$

This can also be written as a matrix product:

$$\langle u, v\rangle = \begin{pmatrix} \overline{u_1} & \ldots & \overline{u_n} \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$$

We can also derive that:

$$\langle \alpha u + \beta v, w\rangle = \overline{\alpha}\langle u, w\rangle + \overline{\beta}\langle v, w\rangle$$

The scalar prtoduct is often written with the Dirac notation $\langle u|v\rangle$, where $\langle u|$ is called "bra" and $|v\rangle$ is called "ket". While $|v\rangle$ represent the vector v, $\langle u|$ represent a map $V \to \mathbb{C}$.

Schwarz inequality

For any vectors $u, v \in V$ that are linearly dependent, we have that:

$$|\langle u|v\rangle| \leq \sqrt{\langle u|u\rangle \langle v|v\rangle}$$

The norm of a vector u is $||u|| = \sqrt{\langle u|u\rangle}$.

Triangle inequality

For any vectors $u, v \in V$, we have that:

$$||u + v|| \leq ||u|| + ||v||$$

Orthogonal vectors

Two vectors $u, v \in V$ are **orthogonal** if:

$$\langle u|v\rangle = 0$$

A set of vectors $\{u_1, \ldots, u_m\}$ is **orthonormal** if:

$$\forall i, j \in \{1, m\} \;\; \langle u|v\rangle = \delta_{ij}$$

where δ_{ij} is called the Kronecker delta and is defined as:

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Given an orthonormal basis $\{e_1, \ldots, e_n\}$ for an n-dimensional vector space V, we can write:

$$\forall v \in V \;\; \exists \alpha_1, \ldots, \alpha_n | v = \sum_{i=1}^{n} \alpha_i e_i$$

We can compute the values α_i as follows:

$$\langle e_j|u\rangle = \left\langle e_j \middle| \sum_{i=1}^{n} \alpha_i e_i \right\rangle = \sum_{i=1}^{n} \alpha_i \langle e_j|e_i\rangle = \sum_{i=1}^{n} \alpha_i \delta_{ij} = \alpha_j$$

In Dirac notation we can write:

$$|u\rangle = \sum_{i=1}^{n} |e_i\rangle \langle e_i|u\rangle \implies \langle v|u\rangle = \sum_{i=1}^{n} \langle v|e_i\rangle \langle e_i|u\rangle$$

Given the above left equation we must have that:

$$\sum_{i=1}^{n} |e_i\rangle \langle e_i| = I$$

4.3 Hilber spaces

Convergence of vector sequence

A vector sequence $v_m \in V$ **converges strongly** to $v \in V$, written $v_m \to v$ if:

$$\lim_{m \to +\infty} ||v - v_m|| = 0$$

If $v \to v_m$ then:

- Weak convergence: $\lim_{m\to+\infty} ||v_m|| = ||v||$
- Scalar product continous: $\langle u|v\rangle = \lim_{m\to+\infty} \langle u|v_m\rangle$

Cauchy sequence

A vector sequence $v_i \in V$ is a **Cauchy sequence** if:

$$\forall \epsilon > 0 \ \exists n_\epsilon | \forall n, m > n_\epsilon \ ||v_n - v_m|| < \epsilon$$

Hilbert space

A Hilbert space $\mathcal{H}$ is a complete scalar product space, so every Cauchy sequence converges strongly to an element in the space.
Finite-dimensional vector spaces are always complete.

Hilbert spaces also exists for infinite dimensional spaces but for quantum computing only the finite ones are needed.

Topologically closed subspaces

A linear subspace W of a Hilbert space $\mathcal{H}$ is **topologically closed** if any sequence of vectors in W converges in W.

Linear operator

Given an Hilber space $\mathcal{H}$ a linear **operator** is a linear function $A : \mathcal{H} \to \mathcal{H}$ that has two properties:

- Operator sum: $\forall v \in \mathcal{H} \ (A + B)v = Av + Bv$
- Operator product: $\forall v \in \mathcal{H} \ (AB)v = A(Bv)$

For operators A, B and $\lambda \in \mathbb{C}$, we have:

1. $(AB)^\dagger = B^\dagger A^\dagger$
2. $(\lambda A)^\dagger = \overline{\lambda} A^\dagger$
3. $(A^\dagger)^\dagger = A$

The adjoint of an operator A is the operator $A^\dagger$ defined as:

$$\forall u, v \in \mathcal{H} \ \left\langle u \middle| A^\dagger v \right\rangle = \langle Au | v \rangle$$

An operator is called **self-adjoint** (or Hermitian) if $A = A^\dagger$. If an operator is self-adjoint we have:

$$\langle u | Av \rangle = \langle Au | v \rangle = \overline{\langle v | Au \rangle}$$

Any linear operator A on a scalar product vector space with orthonormal basis can be represented in matrix form where:

$$A_{ij} = \langle e_i | A e_j \rangle \qquad A^\dagger_{ij} = \overline{A_{ij}}$$

Unitary operators

We can define an operator as unitary in two ways:

- An operator U is unitary if satisfies:

$$UU^\dagger = U^\dagger U = I$$

- An operator U is unitary if satisfies:

  1. U is surjective
  2. $\forall x \in \mathcal{H}\ ||Ux|| = ||x||$

An operator is **normal** if:

$$A^\dagger A = AA^\dagger$$

Both self-adjoint and unitary operators are normal.

Eigenvalues of operators

An eigenvalue λ of an operator A is d-fold generative if there are d linearly independent eigenvectors associated with λ.
The eigenvectors of a given eigenvalue form a linear subspace, called **eigenspace**.

In Dirac notation if λ is a non-degenerate eigrnvalue of A, we write that the only eigenvector associated to λ as $|\lambda\rangle$ and we have that:

$$A\,|\lambda\rangle = \lambda\,|\lambda\rangle$$

From the equation above we can derive:

$$\langle\lambda|\,A^\dagger = \overline{\lambda}\,\langle\lambda|$$

Properties of eigenvalues of operators

Eigenvalues for self-adjoint and unitary operators have some properties:

1. Eigenvalues of self-adjoint operators are real numbers.
2. Eigenvalues of unitary operators are complex numbers of modulus 1.
3. Eigenvectors of self-adjoint and unitary matrices associated to different eigenvectors are orthogonal.

4.3.1 Spectral Theory

Spectral theorem

The set of all eigenvectors u_{ij} of a normal operator is an orthogonal basis for $\mathcal{H}$, so:

$$\forall v \in \mathcal{H}\ v = \sum_{i=1}^{m}\sum_{j=1}^{d_i} \alpha_{ij} u_{ij} \qquad \alpha_{ij} = \langle u_{ij}|v\rangle$$

where m is the number of eigenvalues and d_i is the degenerate of the i-th eigenvalue.

We have that the dimension of $\mathcal{H}$ is:

$$\dim \mathcal{H} = \sum_{i=1}^{m} d_i$$

In Dirac notation we can write:

$$|v\rangle = \sum_{i=1}^{m}\sum_{j=1}^{d_i} \langle u_{ij}|v\rangle\, |\lambda_{ij}\rangle$$

where λ_{ij} is the j-th eigenvector associated to the i-th eigenvalue.

4.4 Projectors

Orthogonal subspaces

Two linear subspaces $V, W \subset \mathcal{H}$ are orthogonal if every vector of V is orthogonal to every vector of W.
The orthogonal complement of V is:

$$V^{\perp} = \{u \in \mathcal{H} | \forall v \in V\ \ \langle u|v\rangle = 0\}$$

For finite-dimensional Hilbert spaces $(V^{\perp})^{\perp} = V$.

Given a closed subspace $W \subset \mathcal{H}$ we want to write any vector $v \in \mathcal{H}$ as a sum $v_W + v_{W^{\perp}}$ where $v_W \in W, v_{W^{\perp}} \in W^{\perp}$.

Projectors

Let f_i be an orthogonal basis for W, we define:

$$\forall v \in \mathcal{H}\ v_w = \sum_i \langle f_i|v\rangle\, f_i \qquad v_{W^{\perp}} = v - v_W$$

The sum is called the **projector operator** on W, a map $P_W : \mathcal{H} \to W$ defined as $P_W(v) = v_W$. The projector on the orthogonal complement of W is $P_{W^{\perp}} = I - P_W$.

For a projector we have that:

$$P_W(v) = \begin{cases} v & v \in W \\ 0 & v \in W^{\perp} \end{cases}$$

So P_W has only two eigenvalues: 0 and 1.
Two projectors P, Q are orthogonal if and only if:

$$PQ = QP = 0$$

If $P \perp Q$, then $P + Q$ is a projector.
Given a family of projectors P_i such that $P_i P_j = \delta_{ij}$, then:

$$\sum_i P_i = I$$

4.5 Tendor product

Tensor product

Given two n-dimensional vectors u, v, the tensor product is a n^2-dimensional vector w defined by:

$$w_i = u_{i \text{ div } n} \cdot v_{i \mod n}$$

where i div n is the integer division.

The scalar product of tensor vectors is defined by:

$$\langle u \otimes v | w \otimes z \rangle = \langle u | v \rangle \langle v | z \rangle$$

The tensor product of operators A, B is defined by:

$$(A \otimes B) u \otimes v = Au \otimes Bv$$

For suitably sized matrices or operators L, M, N, P we have that:

$$(M \cdot N) \otimes (L \cdot P) = (M \otimes L) \cdot (N \otimes P)$$

This equation can be pictured as a quantum circuit:

